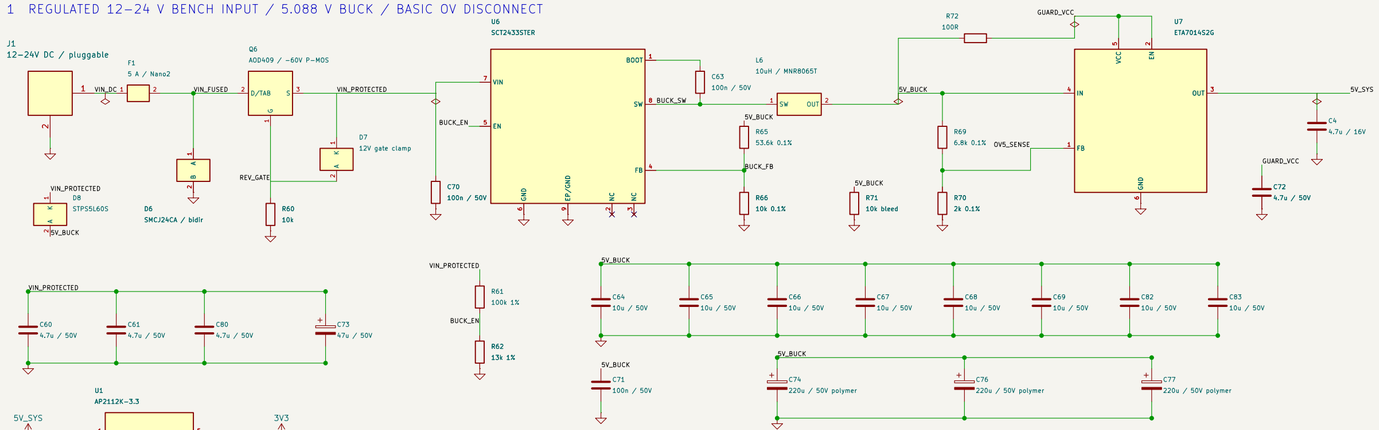
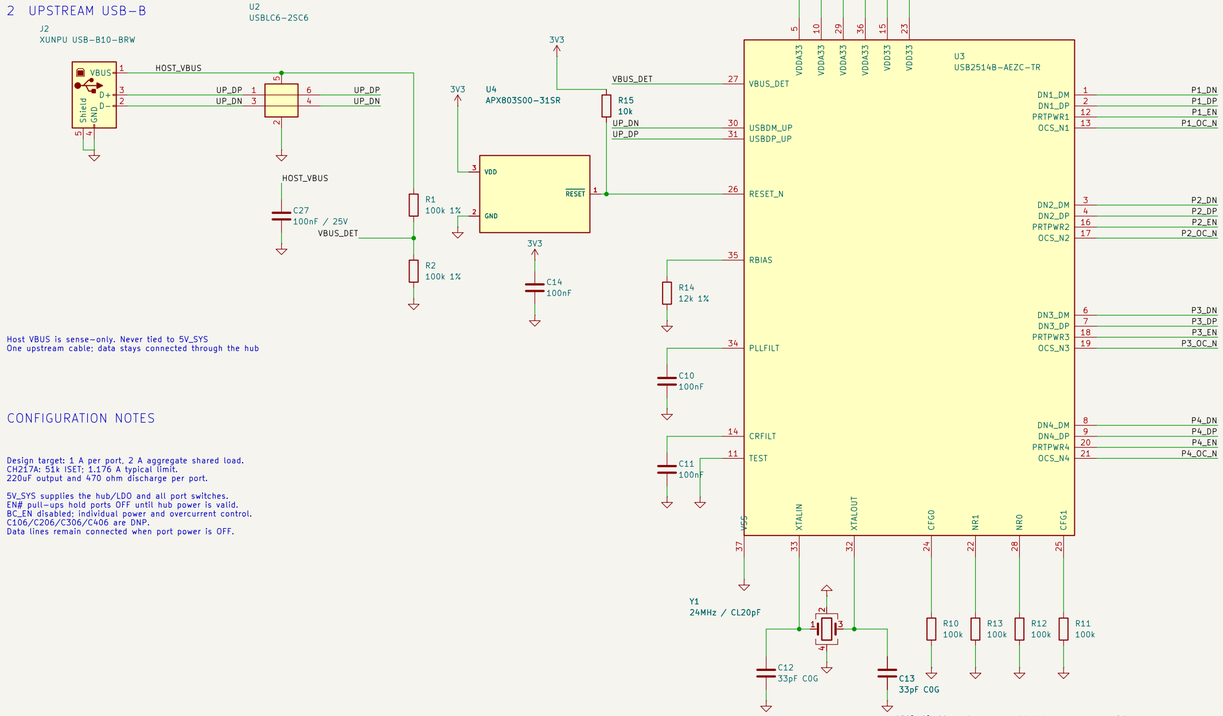
DIN USB DUT HUB | ELECTRICAL SCHEMATIC

USB 2.0 high-speed / self-powered / A8-R5 prototype

1 REGULATED 12-24 V BENCH INPUT / 5.088 V BUCK / BASIC OV DISCONNECT



2 UPSTREAM USB-B



Host VBUS is sense-only. Never tied to 5V_SYS
One upstream cable; data stays connected through the hub

CONFIGURATION NOTES

Design target: 1 A per port, 2 A aggregate shared load.
CH217A: 51k ISET: 1.176 A typical limit.
220uF output and 470 ohm discharge per port.

5V_SYS supplies the hub/LDO and all port switches.
EN# pull-ups hold ports OFF until hub power is valid.
BC_EN disabled: individual power and overcurrent control.
C106/C206/C306/C406 are DNP.
Data lines remain connected when port power is OFF.

ETA7014: independent overvoltage disconnect.
Static trip design range: approximately 5.30-5.79 V.
Not a hard dynamic voltage clamp.

OUTPUT FILTER
8 x 10uF/50V ceramic + 3 x 220uF/50V polymer raw bank.
Ceramic bank design target: >=44uF effective at 5.1 V.
Nominal regulated output: 5.088 V.

GROUNDING: continuous GND reference; no ground split or bead.
Close VIN-GND hot loop locally; Kelvin FB lower resistor to IC GND.
Keep SW/BOOT tight, away from USB and crystal.

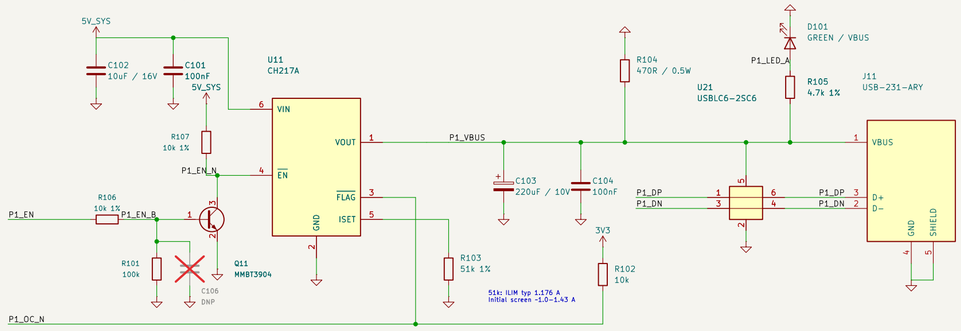
3 HUB CORE AND LOCAL SUPPORT

CFG[1:0]=00: self-powered / individual power and OC
NON_REM[1:0]=00: all ports removable; BC disabled

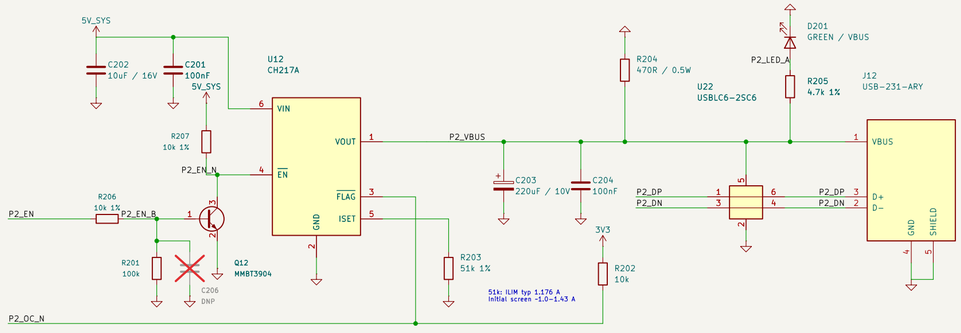
YXC 24 MHz / CL20pF: 33pF loads assume ~3.5pF effective stray.

DIN USB DUT HUB / A8-R5
95 x 80 mm PCB; USB2514B hub.
1 A per port / 2 A shared DUT design target.
Prototype.

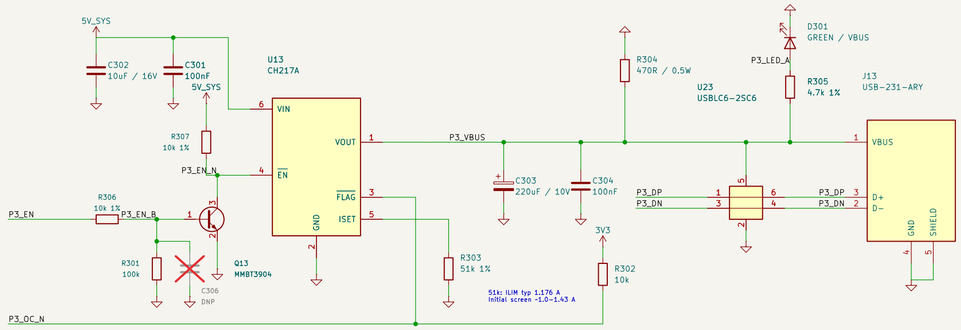
4 PORT 1 - 1 A TARGET / SWITCHED VBUS + USB DATA



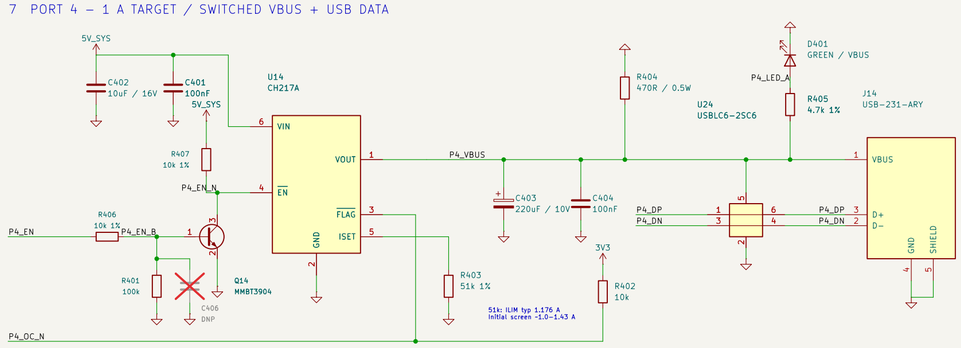
5 PORT 2 - 1 A TARGET / SWITCHED VBUS + USB DATA



6 PORT 3 - 1 A TARGET / SWITCHED VBUS + USB DATA



7 PORT 4 - 1 A TARGET / SWITCHED VBUS + USB DATA



C106/C206/C306/C406: DNP 0805 timing-capacitor sites.
Leave unpopulated; fitted capacitance changes enable/disable timing.

TP1 VIN_DC · TP2 VIN_PROTECTED · TP3 5V_BUCK · TP4 5V_SYS · TP5 OVS_SENSE · TP6 BUCK_FB · TP7 GND · TP8 GND · TP9 GND · TP10 3V3 · TP11 GND

PROTOTYPE / A8-R5 / 1 A PER PORT, 2 A SHARED DESIGN TARGET

Sheet: /
Drawing: Electrical schematic

Title: DIN USB DUT hub electrical schematic

Size: A2	Date: 2026-10-03	Rev: A8-R5-LCSC
RELEASE DRAWING		Id: 1/1